

hospital's intensive care.

F.1: Aim

- To merge the work procedures in the hospital's intensive care units.
- Early examination of malnutrition cases and complications.
- Applying the latest ways of nutritional care for patients in collaboration with other medical staff as well as providing a healthy diet that will help in their fast recovery.
- Contribute to developing ways and to increase efficiency level in curing patients who suffers from food-related diseases.
- Ensuring that patients under hypnosis have specific and appropriate diet according to the nutritional and medical case presented by the physician and Clinical nutrition specialist.
- Modify the style of behavior and spread nutritional awareness for patient's family and society members to prevent nutrition-related problems.
- To share information with medical staff about the importance of nutritional care and its role in the treatment of diseases which can help to elevate their knowledge
- Coordination with authorized staffs in analyzing the effects of such cases.

F.2: Policies:

- Screening of all hypnotic patients in the hospital are according to the system, rules and programs of the ministry of health associated with clinical guidance work.
- Patients who have been identified at high risk must follow specific and documented procedures to evaluate.
- The nutritional assessment should be done by the Clinical nutrition specialist or physician using examination paraphernalia for any patient on high risks within 24 hours since the hypnosis in the intensive care unit.
- Patients must be reassessing every 3 to 5 days or daily if it is necessary or as needed.

- Long-term patients (resident at the hospital for more than a month) with stable condition on oral feeding or organized tubal/intravenous nutrition are being reassess once a month or as required by the patient.
- Patients under tube feeding for more than five days are being observed by NOD with documented information within 24 hours.
- Food intake must be documented within 24 hours in the nutritional assessment form provided on the patient's file until the clinical nutrition specialist submitted the calculated and required calories, fats, protein, carbohydrates.
- The recommendations of clinical nutrition specialist (if it is necessary) may include the following:
 - 1-give dietary supplements according to the patient's condition.
 - 2-tube feedings.
 - 3-coordination with the pharmacist, doctor and intravenous nutrition staff in case of venous nutrition.
 - 4-Give additives or supplements according to the nutritional status of the patient.
- If the clinical nutritionist is unable to do the first assessment or re-assessment during the specific time because the patient is on treatment or because of an unexpected circumstances the time is extended to an extra 24 hours for evaluation in addition of recommendations about their diet.
- Doctors may consult clinical nutritionist through writing a request of consultation in the referral form.

F.3: Procedures:

- Interview the patient about personal data and other objectives:
 - 1-24 hrs. recall
 - 2-change in weight.
 - 3-taken food (if found).
 - 4-appetite and way of intake (by oral, tube, venous).
 - 5-food allergies.

6-eating and drinking difficulties.

7-other problems (diarrhea, gas, constipation....).

8-determine the physical activity type.

9-laboratory results.

10-chart for children/newborns and physical standards for adults.

- **Determine the nutritional status or risk level.**
- **Determine the food and interactions (if any)**
- **Determine the nutritional needs (calculate the calories, protein, fats and carbohydrates)**
- **Implement a nutritional care plan**
- **provide nutritional care plan including Clinical nutrition**
- **provide nutritional preparation according to medical case to improve Clinical nutrition plan.**
- **Educate conscious and oriented patient, families and other members of medical care staff.**
- **Re-assessment assess and document the patient's response on Clinical nutritional system.**
- **Reviewing all the recommendations of clinical nutritionist and plan an appropriate procedure.**

F.4: plan :

The aim of nutritional support is to prevent and treat both macro- and micronutrient deficiencies. There is now evidence that

nutritional support can reduce length of stay and the incidence of complications in patents in intensive care unit.

Assessment of nutritional status

Body Mass Index (mass [kg] / height [m] ²) is a frequently used tool (with a BMI < 18.5 classed as underweight) and has been shown to be an independent predictor of mortality in seriously ill patients. Nevertheless, it does not reflect the acute changes in nutritional status important in critical illness and is used most for the assessment of

long-term health risks of obesity. Probably the most useful measure of nutritional status is a targeted history and examination. One such method which is widely accepted is known as the Subjective Global Assessment.

Subjective Global Assessment

This is a structured approach to taking a history that includes the following categories:

- 1. Weight change - both chronic (over 6 months) and acute (over 2 weeks).**
 - 2. Changes in food intake.**
 - 3. Gastrointestinal symptoms- nausea, vomiting, diarrhea and anorexia**
 - 4. Functional impairment and is combined with a physical examination looking for evidence of the following:**
 - 1. Loss of subcutaneous fat- looking especially at the chest and triceps.**
 - 2. Muscle wasting- looking especially at temporal region, deltoids & gluteal**
 - 3. Oedema**
 - 4. Ascites**
- As the name implies it is a subjective tool, however it has been shown to be reproducible and also to correlate with mortality in a variety of conditions. Assessment of nutritional status should be used to identify patients at risk of refeeding syndrome.**

Nutritional requirements

To actually measure energy use requires sophisticated equipment so requirements are more often estimated using formulae. One such formula is the Harris Benedict Equation which estimates basal metabolic rate (BMR) in kcal/day.

Harris Benedict Equation:

For Male: $BMR = 13.75 \times \text{weight (kg)} + 5 \times \text{height (cm)} - 6.78 \times \text{age (years)} + 66$

For Female: $BMR = 9.56 \times \text{weight (kg)} + 1.85 \times \text{height (cm)} - 4.68 \times \text{age (years)} + 655$

This will usually give a result of around 25 kcal/kg/day. The equation estimates BMR in afebrile healthy individuals and will therefore need to be modified according to the situation to calculate resting energy expenditure (REE).

Fever increase by 10% for each 1°C above 37 (up to max of 40°C)

Sepsis increase by 9% regardless of temperature

Surgery increase by 6% if patient has had surgery or trauma

Burns Increase by 100% if any size over 30%

A careful balance of macronutrients (protein, lipid and carbohydrate) provides the energy requirements whilst micronutrients (vitamins and minerals) are required in very small amounts to maintain health but not to provide energy.

Protein around 1.5 g/kg/day (range 1.2 to 2.0 g/kg/day for ICU patients) Use 2g/kg/day if severely catabolic e.g. severe sepsis/burns/trauma Provides 5.3 kcal/g.

Lipid Provides 9.3 kcal/g Calories from lipid should be limited to 40% of total calories

Carbohydrate Provides 3.75 kcal/g in vivo Give the remaining the energy requirements as carbohydrate

Route of nutrition:

Nutritional support can be given through one of three routes:

1. Oral
2. Enteral- via a tube directly into gastrointestinal tract
3. Parenteral- intravenous (via either peripheral or central vein)

If at all possible oral or enteral are the preferred routes for nutritional support. They are more physiological, reduce the risk of peptic ulceration, minimize mucosal atrophy (food in the gut lumen is a potent stimulus for mucosal cell growth) and may reduce translocation of bacteria from the intestinal lumen. Situations previously thought to preclude enteral nutrition, including major gastrointestinal surgery or acute pancreatitis, have now been shown to be best treated with enteral nutrition.

1. Oral

If the patient can eat then they should be encouraged to do so, ideally by giving them food they enjoy and helping them if necessary. It is important to know how much the patient is eating to see whether they are receiving adequate nutrition. If not, they will need supplementation either orally or enterally.

2. Enteral The various methods of enteral feeding are discussed below:

- I. Nasogastric-** this is the most common method of feeding in Intensive Care. Potential problems include malposition, difficulty swallowing or coughing, discomfort, sinusitis and nasal tissue.
- ii. Oral tubes–** not suitable for awake patients however should be considered in intubated patients to reduce sinusitis (a risk factor for ventilator-associated pneumonia).
- iii. Enterostomy–** gastrostomy or jejunostomy and can be placed at the time of surgery or as a separate procedure. Once inserted they are well tolerated, however there are risks associated with insertion, displacement and infection (including peritonitis). Evidence from non-ICU patients suggests that there are significant benefits for those who require nutritional support for over 4 weeks.
- iv. Post-pyloric feeding–** nasojejunal or jejunostomy. A nasojejunal tube should be over 120cm long to ensure correct placement. Feeding directly into small bowel avoids the problem of gastroparesis. Small bowel ileus is much less common than gastric ileus and tends to be less prolonged; the small bowel recovers normal function 4-8 hours post laparotomy. Post-pyloric feeding is recommended for patients at high risk of aspiration, those undergoing major intra-abdominal surgery and patients who are intolerant of gastric feeding.

3. Parenteral

Parenteral nutrition is most often given via a central vein as the solutions are usually hypertonic. Preparations for peripheral use are available however they have to be isotonic which means that very large volumes would have to be given to provide adequate nutritional support. This severely limits their usefulness.

Enteral Nutrition

- i. Confirm tube position:**

Tube position must be confirmed both clinically and radiographically if possible. Otherwise it is easy to administer feed directly into the lungs.

ii. Secure tube well

There should always be a high index of suspicion that the tube may have become dislodged.

iii. Sit patient up:

The patient should be sat up to an angle of at least 30° to minimise the risk of reflux and subsequent aspiration of gastric contents. This can still occur around a cuffed tracheal tube.

iv. Start feeding early

Early feeding (within 24 hours to 48 hours of admission or surgery) has been shown to reduce septic complications, length of hospital stays and readmission rates in patients after both upper and lower gastrointestinal surgery. Start EN at the rate 30ml/hr (unless otherwise specified). Increase EN by 10 ml every 4 hours until goal reached (unless otherwise specified).

Hang tube feeding for not more than 6 hours.

Change the bag and tube every 24 hours.

v. Aspirate regularly (e.g. 4 hourly) to ensure that gastric residual volume is less than 200mls. Volumes above this greatly increase the risk of reflux and subsequent pulmonary aspiration and feeding rates should be reduced accordingly. This procedure can be via the feeding tube itself if the distal lumen is in the stomach and if it is wide enough. Otherwise (e.g. if a fine bore tube or if jejunal feeding is being used) a separate gastric tube should be used. Once feeding is established this can stop.

Which enteral feed to use?

To avoid contamination, we tend to use commercially prepared formulas over the prepared in the hospital.

Specific additives

There is a lot of interest in the effects of various additives to nutritional support including:

- 1- Nutrisource: fiber to add an extra fiber to some needed cases.**
- 2- Beneprotein: whey protein designed to mix instantly into a wide variety of foods and beverages without compromising taste or texture. For use with individuals who require additional protein to bolster their regular diet, help promote skin health, wound healing, immune response, and muscle strength.**

Complications of nutritional support:

There are a number of complications it is important to be aware of and to look for in patients receiving nutritional support.

1- Refeeding syndrome:

Refeeding syndrome (RFS) is not a singular condition but a group of biochemical shifts and clinical symptoms that can occur in the malnourished or starved individual upon the reintroduction of oral, enteral or parenteral nutrition.

In the following table you will find the criteria for defining the risk of RFS:

*** At risk:**

- Any patient who has had very little or no food intake for >5 days**

***High risk:**

- Any patient is at higher risk of RFS if they have one or more of the following:**
- BMI < 16 kg/m².**
- Unintentional weight loss >15% within the last 3-6 months.**
- Very little or no nutrition for > 10 days.**
- Low concentrations of potassium, magnesium or phosphate prior to feeding.**

Or if a patient has 2 or more of the following:

- *BMI < 18.5 kg/m².**

- **Unintentional weight loss > 10 % within the last 3-6 months.**
- **Very little or no nutrition for >days.**
- **A history of alcohol abuse or drugs including insulin, chemotherapy, antacids or diuretics.**

Extremely high risk:

- **Patients with a BMI <14kg/m².**

*** Negligible intake for more than 15 days**

To manage RFS:

- **Day 1-3 low risk: 15-25 kcal/ kg/day (40-60% CHO:30-40% Fat)**
- **High risk: 10-15kcal/kg/day.**
- **Extremely high risk: 5-15kcal/kg/day.**
- **Protein: 15-20%**
- **Na <1 mmol/kg for 1-7 days.**
- **K⁺ 1-1.5 mmol/kg**
- **Mg⁺⁺ 0.2-0.4mmol/kg**
- **PO₄ 0.3-0.6 mmol/kg**
- **Fluid 20-30ml/kg/day**
- **Thiamine (200-300mg days 1-5)**
- **Multivitamins during days 1-10 replace specific deficiency of trace elements.**
- **No iron substitution within the first 7 days even if patients have iron deficiency.**

2-overfeeding:

Deliberate overfeeding has been tried in an attempt to reverse catabolism but this does not work and is associated with a poor outcome. It can cause uremia, hyperglycemia, hyperlipidemia, fatty liver (hepatic steatosis), hypercapnia (especially with excess carbohydrates) and fluid overload.

- *Propofol (either 1% or 2%) comes in 10% Intralipid and where it is being used for sedation this must be included in the calculations for nutritional support if over feeding is to be avoided.*
- **Phenytoin:** the feeds should be held for 1–2 hours before and after each phenytoin dose with adequate tube flushes before and after phenytoin administration.

3-Hyperglycemia:

critically ill patients become insulin resistant as part of the stress response. Strict control of blood glucose has been shown to have a profound effect on mortality in a study primarily of post-surgical intensive care patients, reducing ICU mortality by almost half (from 8.0% to 4.6%) as well as reducing in-hospital mortality, length of stay, ventilator days, incidence of septicemia and requirements for both dialysis and blood transfusion. The patients who benefited most were those who stayed on ICU for more than 5 days. Medical patients on ICU for more than 3 days have also been shown to derive some benefit from tight glycemic control however the effects were not so marked and there was an apparent increase in mortality in patients on ICU for less than 3 days. It has been suggested that prevention of hyperglycemia should be the target of treatment in the first three days (glucose < 8.3 mmol/l or 150mg/dl) and then if the patient is still critically ill to then aim for tighter control (4.4 – 6.1 mmol/l or 80 – 110 mg/dl).

- **Aspiration Precautions:**

- 1. Keep the patient head elevated or in reverse Trendelenburg position(if not contraindicated by medical condition).**
- 2. In intubated or tracheostomized patients, check residual volume RV every 4 hours. If RV <500 ml and no other signs of intolerance, return the aspirate and continue as per protocol. If RV >500 ml or 200-500 ml with**

other signs of intolerance, return up to 500 ml. hold feeding for 2 hours then resume feeding at the previous rate.

3. In patients who are at high risk of aspiration such as patients who are not intubated or tracheostomized, check RV every 4 hours, if RV <200 ml, return the aspirate and continue as per protocol; if RV >200, return up to 200 ml. Hold feeding for 2 hours then resume feeding at half of the previous rate.
4. If RV remains high, inform the physician to prescribe motility promoting agents such as prokinetic drugs or divert to the level of feeding by postpyloric tube placement.

Guidelines for NPO

Re-evaluate of NPO orders every 4 hours.

- NPO orders related to surgical procedures:

- a) for non-airway or abdominal surgery (head, chest, and orthopedic procedures) tube feeding should not be stopped in patients with and endotracheal (ET) or tracheostomy tube in place.
- b) For open and percutaneous tracheostomy and bronchoscopy , tube feeding should stopped 4 hours prior to procedure and the nasogastric tube should be aspirated at the time of anesthetic induction. Feeding should be restarted immediately on arrival back to the ICU or after the end of the percutaneous tracheostomy procedure.
- c) For abdominal surgery, tube feeding should be stopped 4 hours prior to the procedure if definitive airway (ET or tracheostomy tube) is in place or for 8 hours if patients is not intubated)

- NPO orders related to radiological investigations:

- a. Abdominal or pelvic ultrasound NPO for 6 hours prior exam.
- b. Abdominal or pelvic CT: NPO for 6 hours prior exam.
- c. GI Series/ small bowel series exam of anastomosis site: NPO for 8 hours prior exam.
- d. MRCP and MRI of the abdomen: NPO for 6 hours before the exam.
- e. Red Blood Cells scan: NPO for 4 hours prior exam.

Tube feeding will be restarted by physician order.

If any of the above examinations is scheduled for next day, the primary nurse will call the radiology department to determine the approximate exam time and accordingly the NPO time. If the exam time is not determined, then NPO order will start at 2:00 AM.

For other radiological exams, including CT of brain and chest, MRI of brain, VQ scan, brain perfusion scan, keep patient NPO when the patient is called to the radiology department, aspirate and discard the gastric content.
- the following are the guidelines for interventional radiology procedures to be performed for ICU patients on tube feeding:

- a. Drainage of intraabdominal collections: NPO for 6 hours prior to procedure.**
- b. Angiography: NPO for 4 hours prior to procedure.**
- c. Insertion of hemodialysis catheter: NPO for 4 hours prior to procedure.**
- d. Drainage of pleural effusion: keep patient NPO when the patient is called to radiology department, aspirate and discard the gastric contents.**
- e. IVC filter placement: keep patient NPO when the patient is called to the radiology department, aspirate and discard the gastric contents.**
- f. Insertion of PICC line: keep the patient NPO when the patient is called to the radiology department, aspirate and discard the gastric contents.**

Tube feeding will be restarted directly after returning to the ICU if there are no signs of gastrointestinal intolerance unless specified to keep NPO by the ICU physician. Tube feeding rate will be adjusted according to feeding guidelines to compensate for the NPO time.

Adjust of tube feeding after NPO

Tube feeding rate needs to be increased after NPO status to optimize the delivery of caloric requirement to our critically ill patients.

Feeding adjustment need to be done as follow:

1. Calculate the volume of feeding missed (in ml) during the NPO time: feeding rate before NPO multiplied by number of NPO hours. This equal to X. $X = \text{pre-NPO feeding rate} \times \text{NPO hours}$.
2. Multiply pre-NPO feeding rate by 0.3. this is equal to Y. $Y = \text{pre-NPO feeding} \times 0.3$.
3. Divide X by Y. this will be Z, which is the number of hours needed for compensation. $Z = X/Y$.
4. Adjust tube feeding rate by adding Y to the pre-NPO rate and continue the adjusted rate for Z hours.

Example: a patient was mad NPO for 5 hours to perform tracheostomy. The pre-NPO tube feeding rate was 75 ml/hour. Tube feeding was resumed directly after proper nasoduodenal tube placement was confirmed. Tube feeding rate should be adjusted as follows:

$X = 75\text{ml} \times 5 = 375\text{ml}$.

$Y = 75 \times 0.3 = 23 \text{ ml}$.

$Z = 375/23 = 16 \text{ hours}$.

Tube feeding rate should be adjusted to $75 + 23 = 98 \text{ ml}$ for 16 hour, after which the rate should be reduced to goal rate.

Note: this does not apply to patients with gastrointestinal intolerance and to patients whose feeding being increased as per feeding protocol.

F.5: Appendices:

- Primary nutritional assessment form of the patient.
- Patient's referral form.
- Nutritional re-assessment form of patient.
- Nutritional examination form.
- 24 hours food intake sheet form
- Statement analysis of pharmaceutical nutritional preparations.
- Growth charts for children and newborns.